Daniel Ernesto Mayo

Senior Infrastructure & Linux Engineer | DevOps & Automation | HPC | Cloud & MLOps

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Senior technology professional with 15+ years of hands-on experience in Linux infrastructure, systems administration, virtualization, networking, automation and scientific computing. Strong production-operations background with Python/Bash automation, Docker, Proxmox, VMware, Slurm and JupyterHub. Currently extending that foundation into Cloud, Platform Engineering, MLOps and AI infrastructure.

CORE EXPERTISE

Linux & Systems
RHEL, CentOS, Ubuntu, Windows Server, hardening, storage, high availability, troubleshooting.

Virtualization
Proxmox, VMware, XenServer, remote desktop environments and mixed physical/virtual infrastructure.

Automation
Python, Bash, Git, repeatable operations, data-processing scripts and service automation.

Containers & Delivery
Docker, Docker Compose, Nginx, reverse proxies, GitHub Actions and CI/CD practices.

HPC & Scientific Computing
Slurm, JupyterHub, Linux clusters, scientific workloads and research-user support.

Cloud & Platform
AWS, Azure and GCP foundations; developing practice with Kubernetes, Terraform and MLOps tooling.

PROFESSIONAL EXPERIENCE

2008-2024

Independent Infrastructure IT & Automation Consultant

Designed, deployed and supported infrastructure, virtualization, networks and automation for corporate and SMB environments.

- Supported environments associated with INVAP, Banco de Córdoba, Tarjeta Naranja and Grupo Edisur.
- Worked across Linux/Windows systems, virtualization, networking, operational troubleshooting and automation.
- Used Python and Bash to reduce repetitive administration and improve reproducibility.

UNC - until 2016

Linux Server Administrator - HPC Infrastructure

Administered Linux services and scientific-computing infrastructure at Universidad Nacional de Córdoba.

- Managed research workloads on the Mendieta and Serafín HPC clusters.
- Worked with Slurm workload management and JupyterHub collaborative environments.
- Supported scientific users, compute resources and Linux-based services.

2024

Lecturer - Python Applied to Chemistry

Designed and delivered Python content for scientific data processing, automation and computational modeling.

- Connected programming practices with reproducible scientific workflows and data analysis.

SELECTED ENGINEERING IMPACT

- Multilingual RAG assistant: Gemini + business knowledge retrieval + Docker service + health checks + n8n/WhatsApp fallback.
- Private application exposure: Ubuntu/Proxmox + Docker Compose + Nginx + Cloudflare Tunnel + DNS/HTTPS, with the application bound locally.
- Trilingual Astro portfolio: EN/ES/□□ routes + GitHub Actions + GitHub Pages automated deployment.

EDUCATION

BSc in Chemistry - Physical Chemistry - Universidad Nacional de Córdoba (2016)

Diploma in Data Science, Artificial Intelligence and Applications - Universidad Nacional de Córdoba (2024)

Higher Technician in Mechatronics - Universidad Tecnológica Nacional (2012)

Linux Server Administrator - Universidad Nacional de Córdoba (2014)

LANGUAGES & TARGET ROLES

Spanish: native | English: advanced B2/C1 | Portuguese: advanced | Mandarin Chinese: intermediate (HSK4)

Target roles: Senior Infrastructure Engineer, Linux Engineer, DevOps/Platform Engineer, Automation Engineer, HPC Engineer, Cloud Engineer, MLOps/AIOps, Data/AI Infrastructure.

SELECTED TECHNICAL STACK

Systems Linux, RHEL, Ubuntu, Windows Server, storage, HA	**Automation** Python, Bash, Git, SQL, repeatable workflows
Platforms Proxmox, VMware, XenServer, Docker, Nginx	**HPC** Slurm, JupyterHub, cluster operations
Cloud path AWS, Azure, GCP; Kubernetes/Terraform in development	**Data/AI** PostgreSQL, MongoDB, RAG, MLflow/MLOps concepts

PROFESSIONAL STRENGTHS

- End-to-end troubleshooting across systems, virtualization, networking and applications.
- Strong operations-to-automation mindset: stabilize first, then make the workflow repeatable.
- Able to connect infrastructure, scientific computing, data and AI teams.
- Multilingual communication for international and distributed environments.

Recruiter note	Open to remote and international opportunities. Portfolio and project case studies: https://shifudem.github.io/danielmayo/